

Name: _____

- There are 12 points possible on this proficiency, one point per problem. **No partial credit will be given.**
- You have one hour to complete this proficiency.
- No aids (book, calculator, etc.) are permitted.
- You do **not** need to simplify your expressions.
- Correct parenthesization is required.
- Do not put a $+C$ where it does not belong, and you must include $+C$ where it is needed.
- You must show sufficient work to justify your final expression. A correct answer for a nontrivial computation with no supporting work will be marked as incorrect.

1. [12 points] Compute the following integrals.

a. $\int_{-1}^e 2x^2 - \frac{3}{x} dx$

b. $\int_1^3 (2x+3)e^{2x^2+6x} dx$

c. $\int x\sqrt[3]{x+2} dx$

d. $\int 7 \sin x + \ln 4 \, dx$

e. $\int x^\pi + \pi \cos x \, dx$

f. $\int 4 \sec^2 x + \frac{1}{x^5} \, dx$

g. $\int e^x \sec(e^x) \tan(e^x) dx$

h. $\int \frac{3}{\sqrt{1-x^2}} dx$

i. $\int l \sin(k\theta) d\theta$ where k and l are positive constants

j. $\int \sin(5)x^{-3} + e \, dx$

k. $\int 4x^{5/3} - \frac{2}{x^{3/5}} \, dx$

l. $\int (x^2 - 4)(x + 5) \, dx$